SCIENCE 1.6 **Name:_____**

WORKSHEET ONE **DISTANCE, TIME, SPEED AND ACCELERATION**

1. Complete the following chart. (A)

Quantity	Quantity symbol	Unit name	Unit symbol
Distance		metre	
	m		
			s
Speed			
	a		ms^{-2}

2. Carry out the following unit conversions. The first one is done for you. (A)

(a) 10m = ***10,000*** mm
(b) 10min = _____ s
(c) 1hour = _____ s
(d) 600g = _____ kg
(e) 2.5km = _____ m
(f) 950m = _____ km
(g) 4.3kg = _____ g
(h) 1.5hours = _____ s
(i) 2.5min = _____ s
(j) 500mm = _____ m

3. (a) An airplane flies 5000 m in 20 s. What is its speed in ms^{-1} ? (M)

(b) A car goes from Nelson to Greymouth (305 km) in 4.5 hours. What is its average speed in kmh^{-1}? (M)

(c) A cyclist travels 1500 m in 4 min. What is his speed in ms^{-1}? (M)

(d) A journey from Napier to Wellington is made in three sections.

> First section: Napier to Woodeville. Distance = 147 km. Time = 1.8 h.
> Second section: Woodeville to Masterton. Distance = 82 km. Time = 0.9 h.
> Third section: Masterton to Wellington. Distance = 102 km. Time = 1.7 h.

Calculate the speed of travel over each stage of the journey and the average speed of the whole journey. (A,M)

4. (a) A seagull flies at 10 ms^{-1} for 5 min. How far does it fly? (M)

(b) A snail crawls a distance of 3.8 cm at a speed of 0.002 ms^{-1}. How long does it take to do this? (M)

(c) The average speed of a journey from Whangarei to Auckland is 84 kmh^{-1}. If the distance travelled is 170 km, how many hours did it take? (M)

5. Use the speed formula to solve and complete the following chart. (v = d/t) (A)

Speed (ms^{-1})	12		50	0.8	5.0	
Distance		500 m	1.4 km		100 m	2 km
Time	11 s	1 min		25 s		10 min

6. A book falls off a shelf and accelerates towards the ground. (A)
What is its acceleration? _____
How fast will it be going after one second? _____

7. A car accelerates from rest and reaches a speed of 15 ms^{-1} after 3 seconds. What is its acceleration? (M)

8. A rocket is fired into the air and its upward speed is 13 ms^{-1} as it clears the top of a nearby building. Three seconds later its upward speed is 25 ms^{-1}. What is its acceleration? (M)

9. A girl, starting from rest, skis down a mountain slope and after six seconds is travelling at a speed of 14 ms^{-1}. What is her acceleration? (M)

10 Using the equation for acceleration complete the following chart. (a = $\triangle$v/t) (A)

Acceleration	2.4 ms^{-2}		0.8 ms^{-2}	18 ms^{-2}		30 ms^{-2}
Change in speed	60 ms^{-1}	44 ms^{-1}		120 ms^{-1}	60 ms^{-1}	
Time		6 s	2 min		15 s	3.5 s

11. A car is travelling at 5 ms^{-1} and accelerates at 4 ms^{-2} for 6 s. What is its final speed? (M)

12. A cyclist travelling at 3 ms^{-1} accelerates at 0.5 ms^{-2}. What will her speed be after 4 s? (M)